

Disproof of conjecture 00000003837

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Verdict: FALSE

We disprove conjecture 00000003837 as stated. The refutation is unconditional and elementary. Its first clause fails in the smallest possible case, type A_1 with $\lambda = \omega_1$: the crystal $B(\omega_1)$ has two elements and the star involution has 0 fixed points, which is even. Its second clause fails there too, because $0 < 3$; and in type A_1 the hypothesis $\lambda = -w_0\lambda$ holds for *every* weight, while the fixed-point count over all of type A_1 is at most 1.

1 The conjecture

The statement, quoted verbatim from `conjectures/00000003837.md`:

English. Definition: The star involution $*$ denotes the crystal involution reversing all simple arrow directions, always existing in finite type. Conjecture: The number of fixed points of $*$ in $B(\lambda)$ is always odd, and when $\lambda = -w_0\lambda$ the number of fixed points is at least 3. (star fixed point parity)

中文. 定义: 星对合 $*$ 指同时反转所有简单箭头方向所得的晶体对合, 有限型上恒存在。猜想: $B(\lambda)$ 中 $*$ 的不动点个数恒为奇数, 且当 $\lambda = -w_0\lambda$ 时不动点个数不少于 3。(星不动点奇偶)

The claim has two clauses.

1. **Parity clause.** For every dominant integral weight λ in finite type, the star involution $*$ on $B(\lambda)$ has an odd number of fixed points.
2. **Three clause.** Whenever $\lambda = -w_0\lambda$, that number of fixed points is at least 3.

Both are false, and both fail in the same minimal example.

2 The elementary parity lemma

Lemma 1 (Parity lemma). *Let σ be an involution of a finite set X . Then*

$$\#\text{Fix}(\sigma) \equiv |X| \pmod{2}.$$

Proof. For every $x \in X \setminus \text{Fix}(\sigma)$ we have $\sigma(x) \neq x$, and $\sigma^2 = \text{id}$ shows that σ maps the two-element set $\{x, \sigma(x)\}$ to itself and exchanges its elements. These 2-cycles partition $X \setminus \text{Fix}(\sigma)$, so $|X| - \#\text{Fix}(\sigma)$ is even. Hence $\#\text{Fix}(\sigma) \equiv |X| \pmod{2}$. $\square$

Since $|B(\lambda)| = \dim V(\lambda)$, the lemma yields the correct general congruence

$$\# \text{Fix}(\ast) \equiv |B(\lambda)| = \dim V(\lambda) \pmod{2}. \quad (1)$$

In particular the parity clause can hold only for those λ with $\dim V(\lambda)$ odd; as an unconditional assertion about all λ it is false. Applied to $B(\omega_1)$, whose cardinality is 2, the lemma already forces $\# \text{Fix}(\ast)$ to be even.

3 The counterexample: type A_1 , $\lambda = \omega_1$

Definition 2. In type A_1 (that is, for $\mathfrak{sl}_2$) let $\lambda = \omega_1$. The crystal is the two-element set

$$B(\omega_1) = \{u_+, u_-\}, \quad \text{wt}(u_+) = +1, \quad \text{wt}(u_-) = -1,$$

with a single arrow

$$f_1(u_+) = u_-, \quad f_1(u_-) = 0, \quad e_1(u_-) = u_+, \quad e_1(u_+) = 0.$$

The star involution $\ast$ reverses all arrows.

Theorem 3. *On $B(\omega_1)$ the arrow-reversing involution is forced to be the swap*

$$\ast(u_+) = u_-, \quad \ast(u_-) = u_+,$$

and it has no fixed point:

$$\# \text{Fix}(\ast) = 0.$$

Consequently $\# \text{Fix}(\ast)$ is even, and $\# \text{Fix}(\ast) < 3$.

Proof. Reversing the unique arrow means $f_1 \circ \ast = \ast \circ e_1$ (with $\ast$ extended by $\ast(0) = 0$). Evaluating at the bottom element u_- , where $e_1(u_-) = u_+$ and f_1 is defined, gives

$$\ast(u_+) = \ast(e_1(u_-)) = f_1(\ast(u_-)),$$

so $\ast(u_+)$ lies in the image of f_1 , namely $\{u_-\}$. Hence $\ast(u_+) = u_-$; applying the involution law $\ast^2 = \text{id}$ gives $\ast(u_-) = u_+$. Both elements are moved, so $\# \text{Fix}(\ast) = 0$. As 0 is even and $0 < 3$, both clauses fail. $\square$

Corollary 4. *The hypothesis of the three clause holds at $\lambda = \omega_1$: in type A_1 the longest element $w_0 = s_1$ acts on the weight lattice by -1 , so $-w_0$ acts as the identity and*

$$-w_0 \omega_1 = \omega_1.$$

Nevertheless $\# \text{Fix}(\ast) = 0 < 3$. Thus the three clause fails although its hypothesis is satisfied, and the parity clause fails at the same time.

Remark 5. Nothing here is a boundary or degeneracy artefact. The weight ω_1 is the simplest non-trivial dominant weight in the simplest finite type, and $B(\omega_1)$ is the simplest non-trivial crystal. Any faithful definition of the star involution must agree with the swap by Theorem 7 below.

4 The general family $B(k\omega_1)$ and the table

For $k \geq 1$ the crystal $B(k\omega_1)$ is the $(k+1)$ -element chain with weights

$$k, k-2, \dots, -k.$$

Index its elements by positions $0, 1, \dots, k$ from the top, so that j has weight $k-2j$. The involution $j \mapsto k-j$ is the unique arrow-reversing involution, and it negates weights. Its fixed points solve $j = k-j$, i.e. $2j = k$: there is exactly one when k is even (the weight-0 element) and none when k is odd. Therefore

$$\# \text{Fix}(\ast) = \begin{cases} 1, & k \text{ even,} \\ 0, & k \text{ odd.} \end{cases} \quad (2)$$

For $k = 1, \dots, 5$:

k	λ	weights of $B(k\omega_1)$	$ B(k\omega_1) $	$\# \text{Fix}(\ast)$	parity	$\# \text{Fix}(\ast) \geq 3?$
1	ω_1	1, -1	2	0	even	no ($0 < 3$)
2	$2\omega_1$	2, 0, -2	3	1	odd	no ($1 < 3$)
3	$3\omega_1$	3, 1, -1, -3	4	0	even	no ($0 < 3$)
4	$4\omega_1$	4, 2, 0, -2, -4	5	1	odd	no ($1 < 3$)
5	$5\omega_1$	5, 3, 1, -1, -3, -5	6	0	even	no ($0 < 3$)

In the $k = 2$ row the crystal has 3 elements (odd cardinality) and the fixed-point count is 1, consistent with (1) and odd as the parity clause would demand — but $1 < 3$. So the failure of the three clause is not a mere parity artefact either.

5 The clause “ $\# \text{Fix}(\ast) \geq 3$ ” fails for every λ in type A_1

By (2), over all $k \geq 1$ the fixed-point count of the star on $B(k\omega_1)$ is either 0 or 1; hence

$$\max_{k \geq 1} \# \text{Fix}(\ast \text{ on } B(k\omega_1)) = 1 < 3.$$

Since in type A_1 one has $-w_0 = \text{id}$, so that $\lambda = -w_0\lambda$ for every dominant weight λ , the three clause *fails for every λ in type A_1* , not only for $\lambda = \omega_1$. The parity clause, meanwhile, holds exactly for the even k (for which $\dim V(k\omega_1) = k+1$ is odd), in accordance with (1); this is a tautology on the restricted set of λ with $\dim V(\lambda)$ odd, not the unconditional claim of the conjecture.

6 Convention robustness

One might try to rescue the conjecture by adopting a different convention for $\ast$ or for weights. This is impossible.

Theorem 6. *Every involution of a two-element set has 0 or 2 fixed points. In particular every involution σ of $B(\omega_1)$ has even $\# \text{Fix}(\sigma)$, and $\# \text{Fix}(\sigma) < 3$.*

Proof. A two-element set is $\{a, b\}$. An involution either fixes both elements (identity, 2 fixed points) or swaps them (no fixed points). Both counts, 0 and 2, are even and less than 3. $\square$

Theorem 7 (Uniqueness of the star). *Let $\sigma : B(\omega_1) \rightarrow B(\omega_1)$ be an involution reversing the arrow, i.e. $f_1 \circ \sigma = \sigma \circ e_1$ (with $\sigma(0) = 0$). Then σ is the swap of Theorem 3. Hence any faithful convention for $*$ agrees with it, and $\# \text{Fix}(*) = 0$ is convention-independent.*

Proof. Since $e_1(u_-) = u_+$ and $\sigma^2 = \text{id}$,

$$\sigma(u_+) = \sigma(e_1(u_-)) = f_1(\sigma(u_-)),$$

so $\sigma(u_+) \in \{u_-\}$, i.e. $\sigma(u_+) = u_-$ and then $\sigma(u_-) = u_+$ by involutivity. $\square$

Proposition 8. *There is no involution of $B(\omega_1)$ that both reverses the arrow and preserves weights. The forced star negates weights, so the weight condition cannot be relaxed to manufacture a fixed point.*

Proof. By Theorem 7 any arrow-reversing involution is the swap, which sends the top element u_+ of weight $+1$ to u_- of weight $-1 \neq +1$. $\square$

Thus the value $\# \text{Fix}(*) = 0$ cannot be evaded by changing the definition of $*$: the arrow-reversing requirement pins it down, and no involution of $B(\omega_1)$ at all has an odd or ≥ 3 fixed-point count.

7 Literature context

The genuine results surrounding the star involution are not parity theorems. The relevant theorem is Stembridge’s and Berenstein–Kirillov’s statement that the number of *self-evacuating tableaux* of a shape equals the number of *domino tableaux* of that shape (J. R. Stembridge, *Duke Math. J.* **82** (1996); A. Berenstein and A. Kirillov, *Discrete Math.* **225** (2000), and references therein). This is a bijective equinumerosity statement joining evacuation and domino tableaux; it contains no assertion that $\# \text{Fix}(*)$ is odd. The correct general congruence in this circle of ideas is the elementary (1),

$$\# \text{Fix}(*) \equiv \dim V(\lambda) \pmod{2},$$

which is a consequence of the 2-cycle pairing of Lemma 1. The conjecture appears to be a garbled rendering of these results: its parity clause conflates “ $\dim V(\lambda)$ is odd” with “always odd”, and its three clause has no counterpart in the literature. We state this only as context; the refutation above is self-contained and does not rely on it.

8 Reproducibility

The checks are carried out by `reproduce.py`, which uses the Python 3 standard library only:

```
$ python3 reproduce.py
... (the parity lemma, the  $B(\omega_1)$  computation, the table for  $k = 1..10$ ,
and the robustness checks)
PASS: all checks verified; conjecture 00000003837 is FALSE.
```

It models $B(k\omega_1)$ for $k = 1, \dots, 10$ as an $(k+1)$ -element chain, verifies that $j \mapsto k-j$ is an arrow-reversing involution negating weights, counts its fixed points, asserts that the count is even for every odd k (in particular that $k = 1$ gives 0), asserts $0 < 3$, verifies the parity lemma $\# \text{Fix} \equiv |B(k\omega_1)| \pmod{2}$, and exhaustively verifies that every involution of the two-element set has 0 or 2 fixed points

and that the arrow-reversing involution of $B(\omega_1)$ is unique. It prints **PASS** and exits 0 exactly when every check holds.

9 Formalisation

The counterexample is formalised in the accompanying Lean 4 project under `lean4/`. It builds with `lake build` and is audited by `lake env lean Check.lean`, using core Lean only (no Mathlib) and no `sorry`. The crystal $B(\omega_1)$ is modelled as `Fin 2` with explicit operators, and the main formal statements are

```
theorem star_reverses_f : ∀ j, f1 (star j) = (e1 j).map star
theorem wt_star : ∀ j, wt (star j) = - wt j
theorem star_fixedCount : fixedCount star = 0
theorem conjecture_00000003837_false :
  fixedCount star = 0 ∧ fixedCount star % 2 = 0
  ∧ fixedCount star < 3
theorem involution_fixedCount_even (σ) (hσ : ∀ x, σ (σ x) = x) :
  fixedCount σ % 2 = 0
theorem arrow_reversing_involution_eq_star (σ) (hσ) (hf) : σ = star
theorem no_weight_preserving_arrow_reversal :
  ¬ ∃ σ, (∀ x, σ (σ x) = x) ∧ (∀ j, wt (σ j) = wt j)
  ∧ (∀ j, f1 (σ j) = (e1 j).map σ)
```

Here `star` is the swap on `Fin 2`, `fixedCount` counts the fixed points over `List.finRange 2`, and the theorems `involution_fixedCount_even` and `arrow_reversing_involution_eq_star` formalise the convention-robustness argument. No `sorryAx` occurs; the only axioms reported are `propext` and `Quot.sound`. See `lean4/README.md` for the full statement table and the faithfulness note.